

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

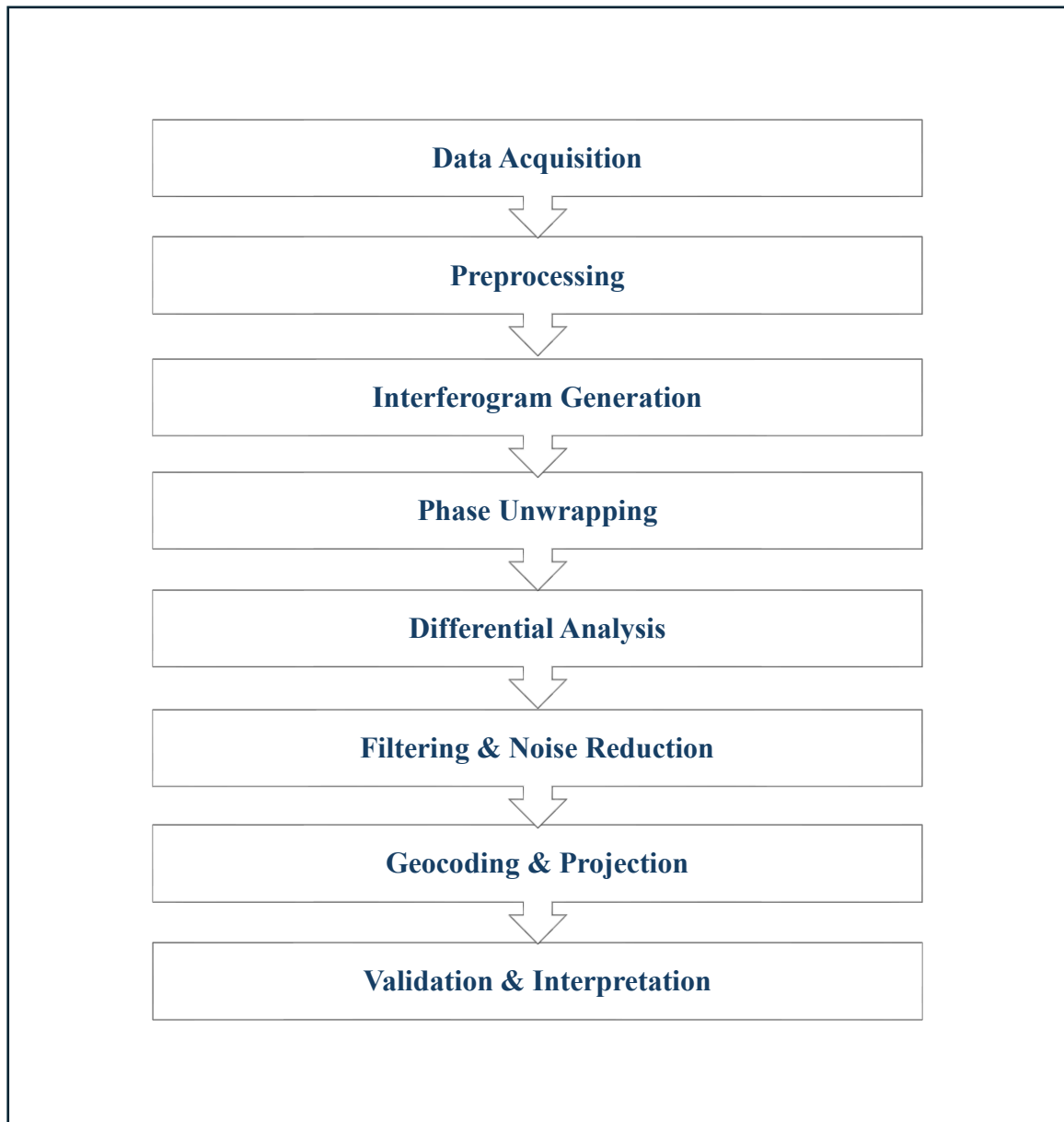
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
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S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
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S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 903/2

Bridge Chainage – 943+240

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MJB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

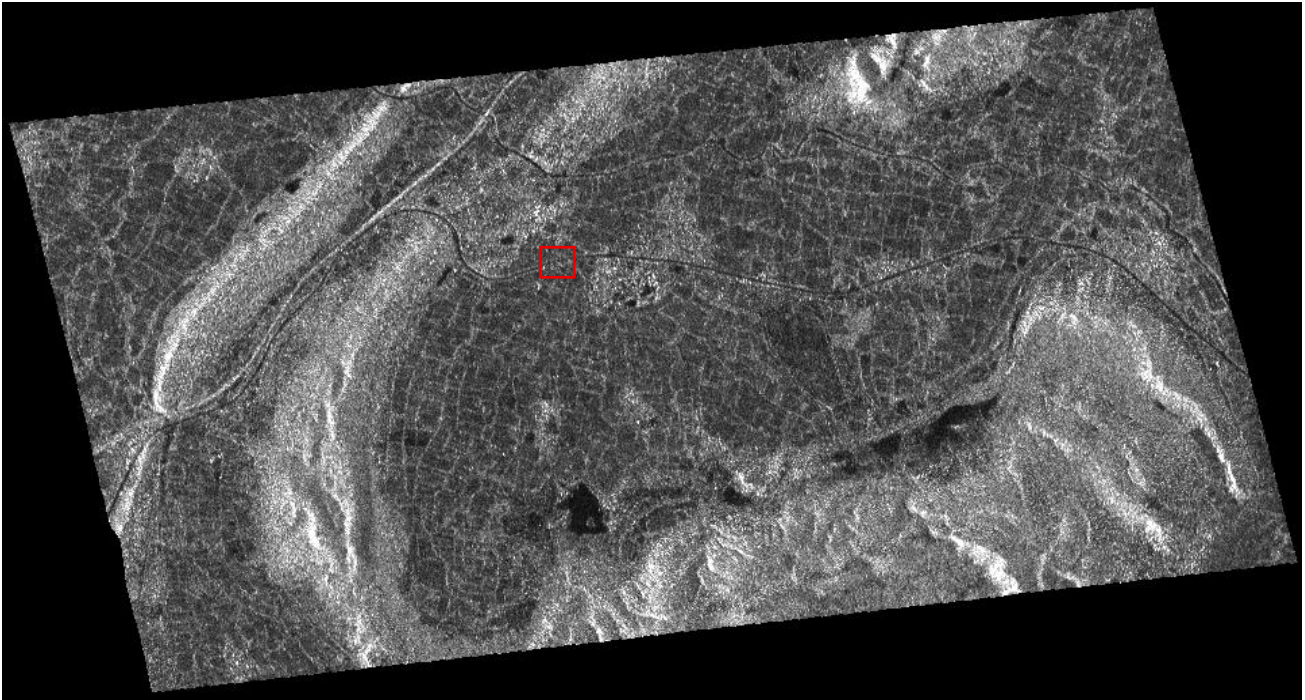
Flight Direction - DESCENDING

Polarization – VV+VH

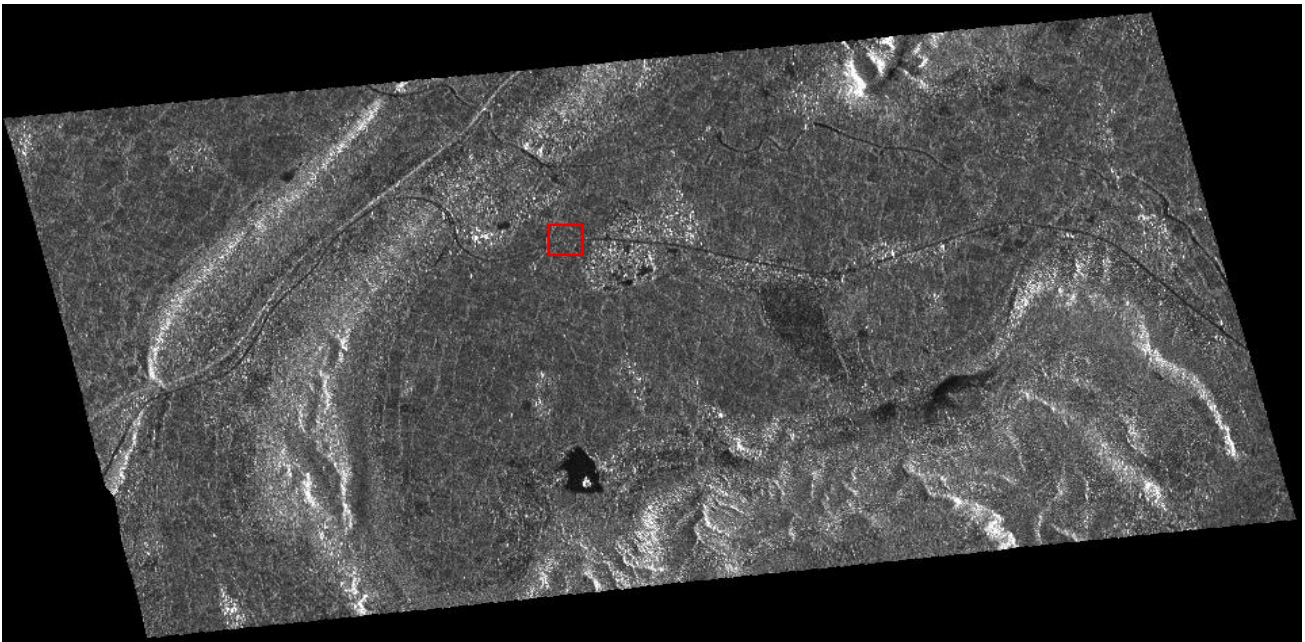
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

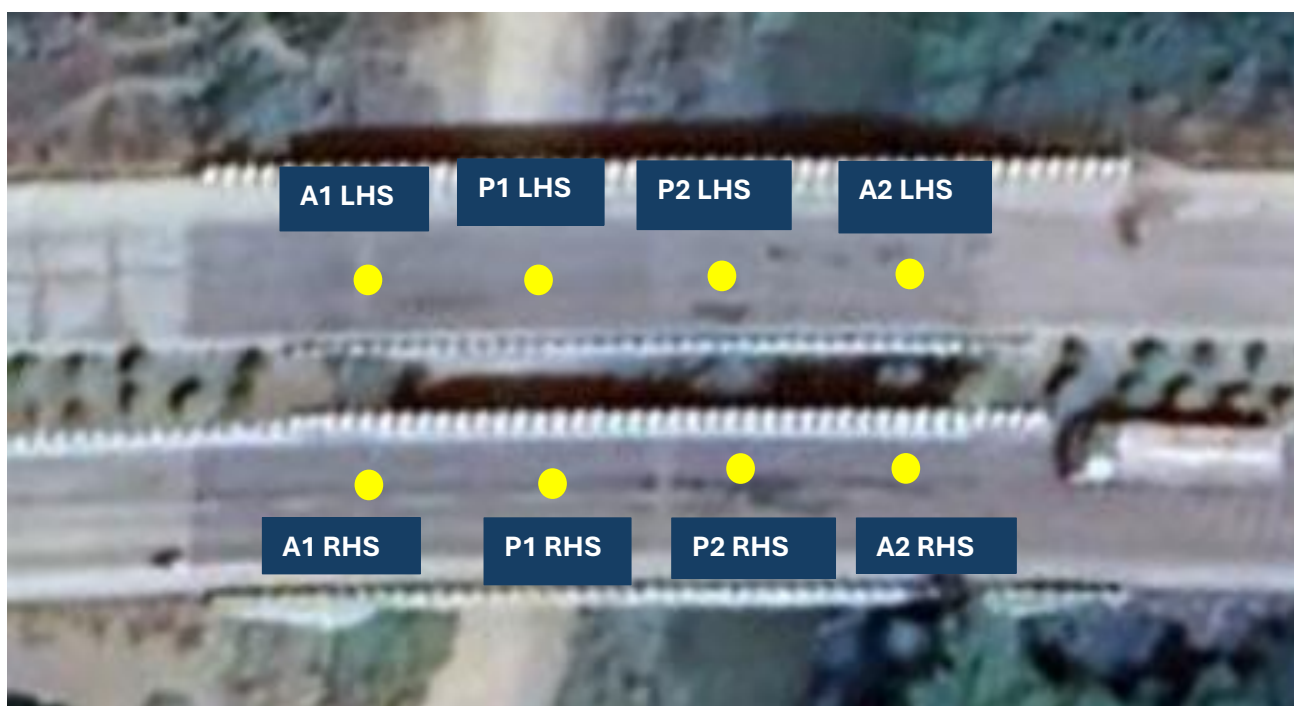


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

MNB – CH-943+240

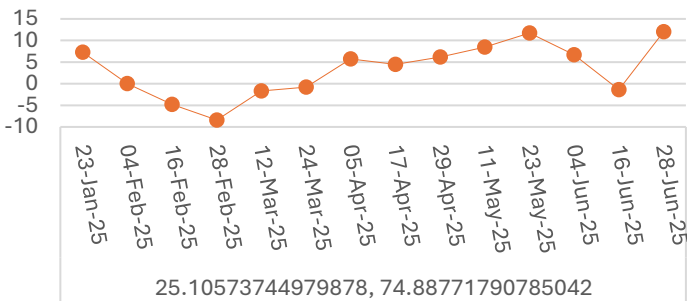
Length – 49.46M



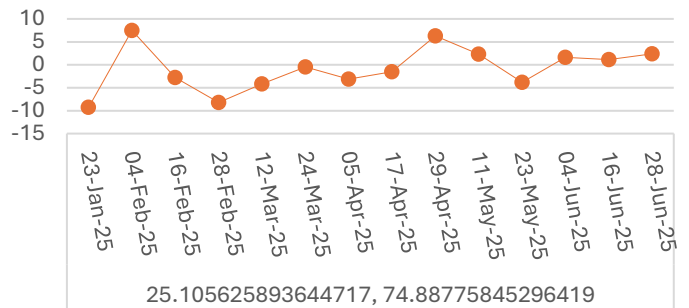
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
P1	Pier 1		
P2	Pier 2		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

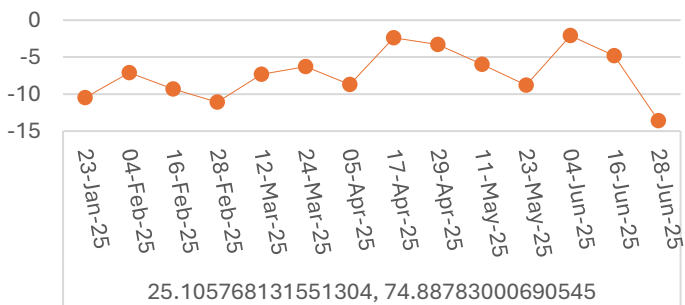
Displacement @ A1 LHS VV



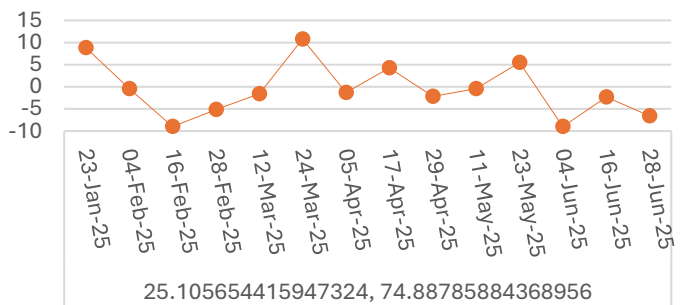
Displacement @ A1 RHS VV



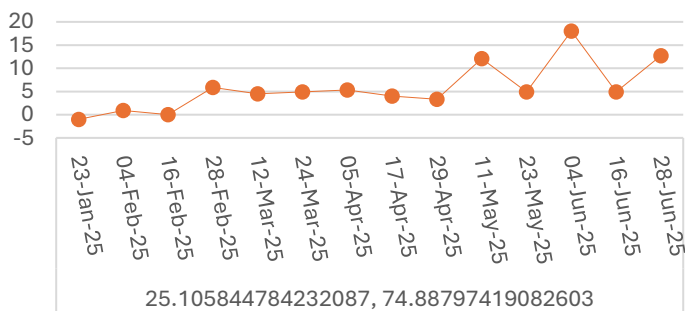
Displacement @ P1 LHS VV



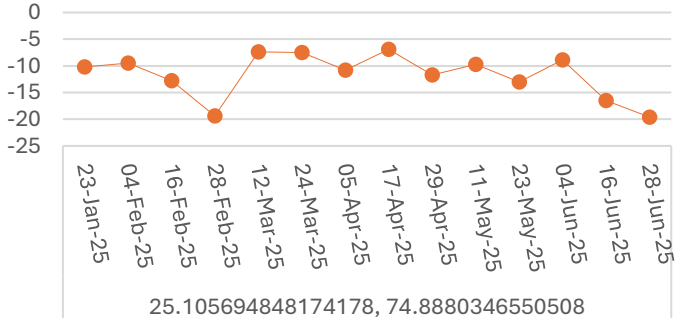
Displacement @ P1 RHS VV



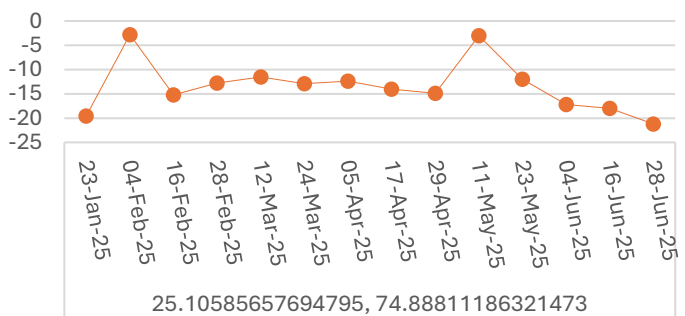
Displacement @ P2 LHS VV



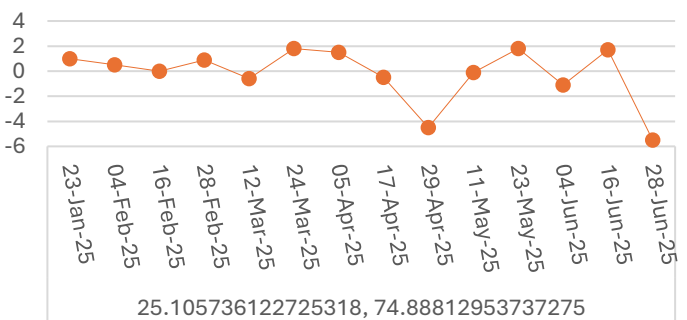
Displacement @ P2 RHS VV



Displacement @ A2 LHS VV

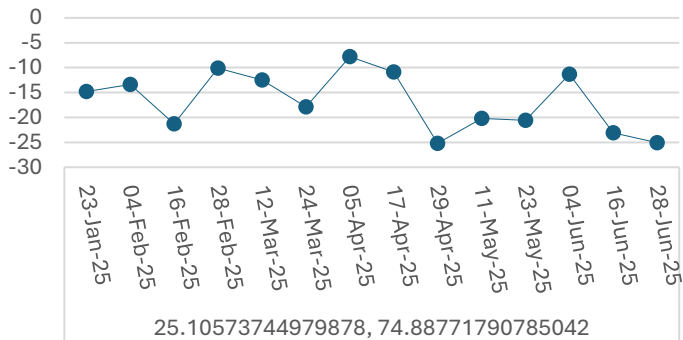


Displacement @ A2 RHS VV

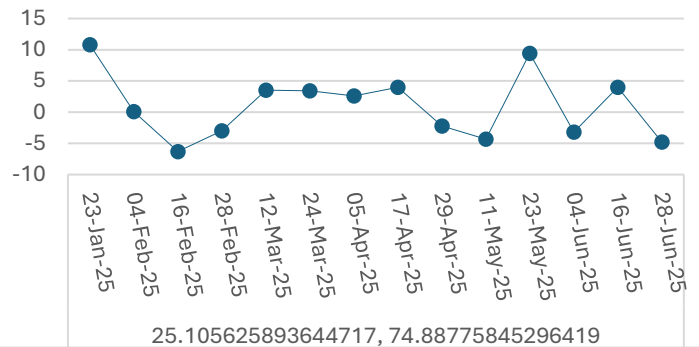


3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

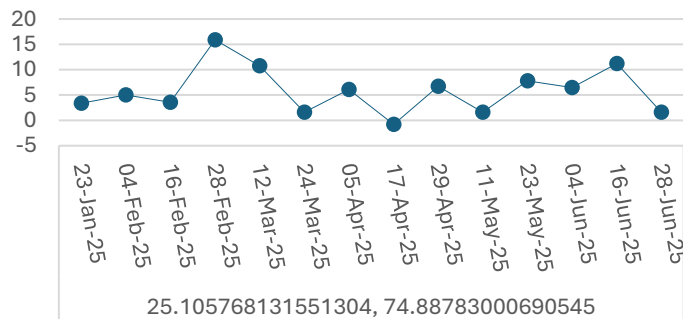
Displacement @ A1 LHS VH



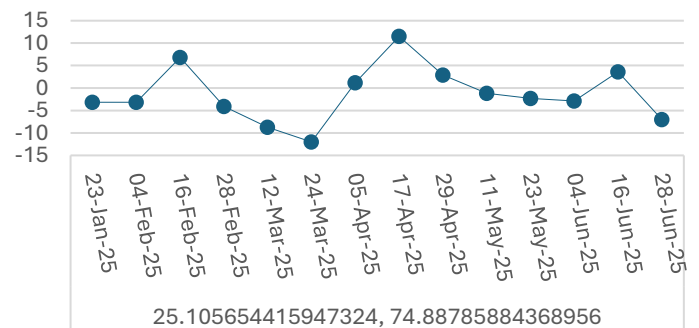
Displacement @ A1 RHS VH



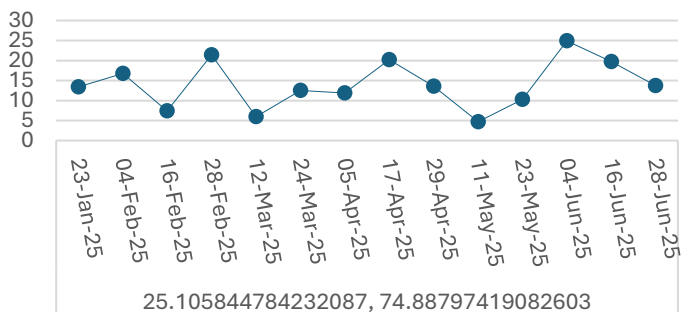
Displacement @ P1 LHS VH



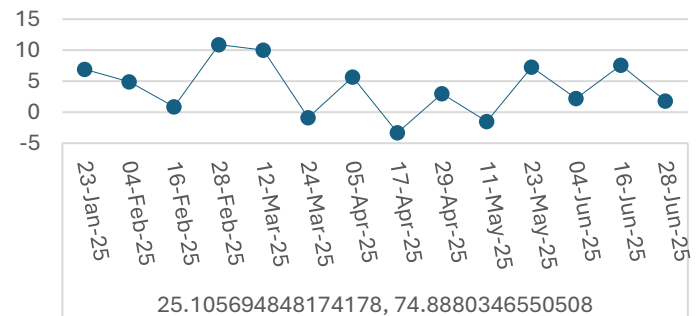
Displacement @ P1 RHS VH



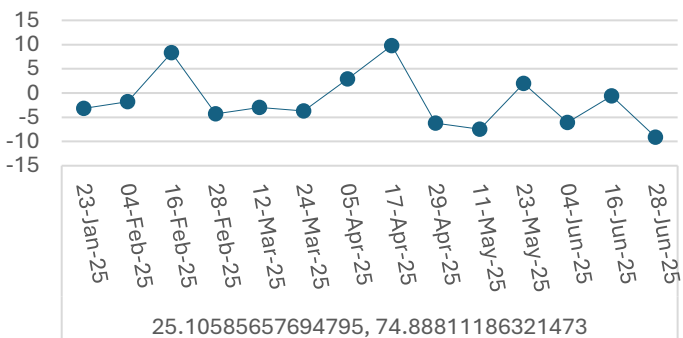
Displacement @ P2 LHS VH



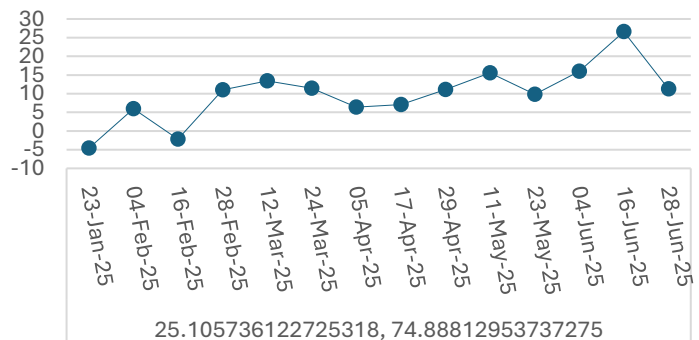
Displacement @ P2 RHS VH



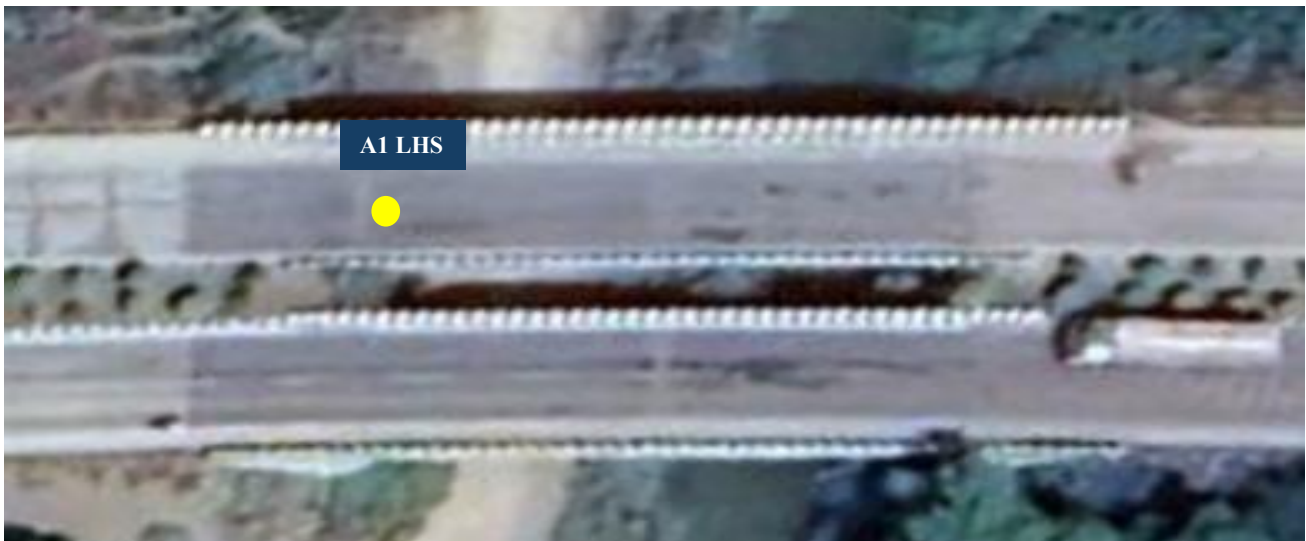
Displacement @ A2 LHS VH



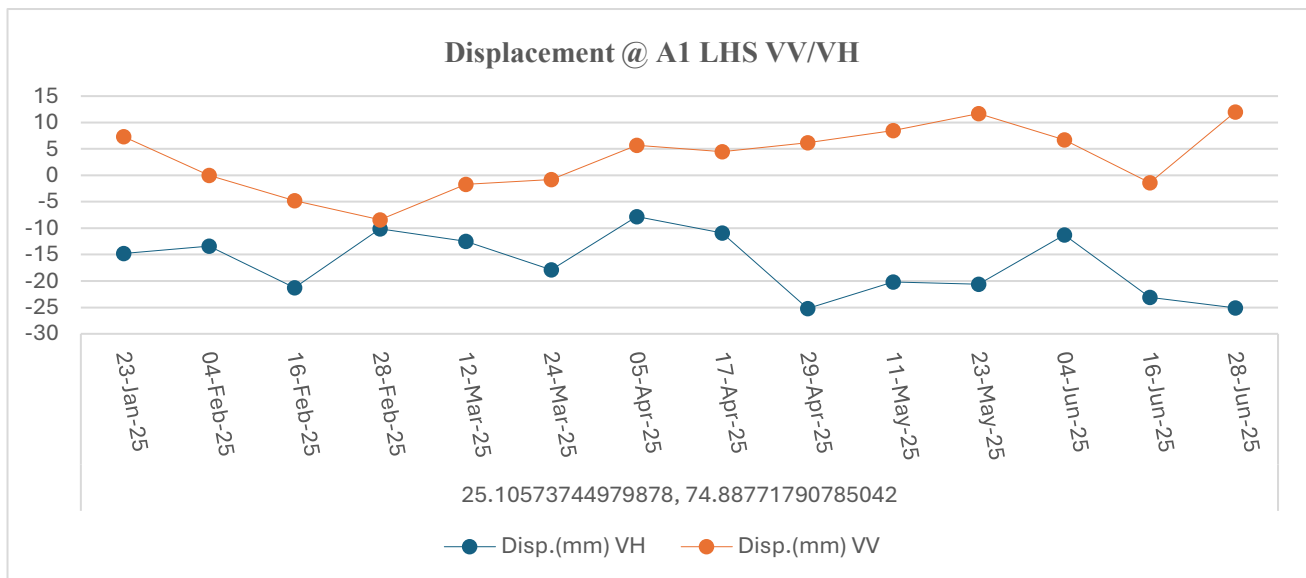
Displacement @ A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.10573744979878, 74.88771790785042)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** displays a general upward trend, rising from around +10 mm in late January 2025 to nearly +15 mm by late June 2025, with minor variations between months. This indicates a progressive vertical uplift at the LHS location.
- **Horizontal displacement (VH)** shows considerable fluctuations over the monitoring period, ranging from approximately –25 mm to –5 mm, indicating a persistent negative lateral shift with several intermittent peaks and dips. The overall behaviour reflects irregular but consistently negative horizontal movement at A1 LHS.

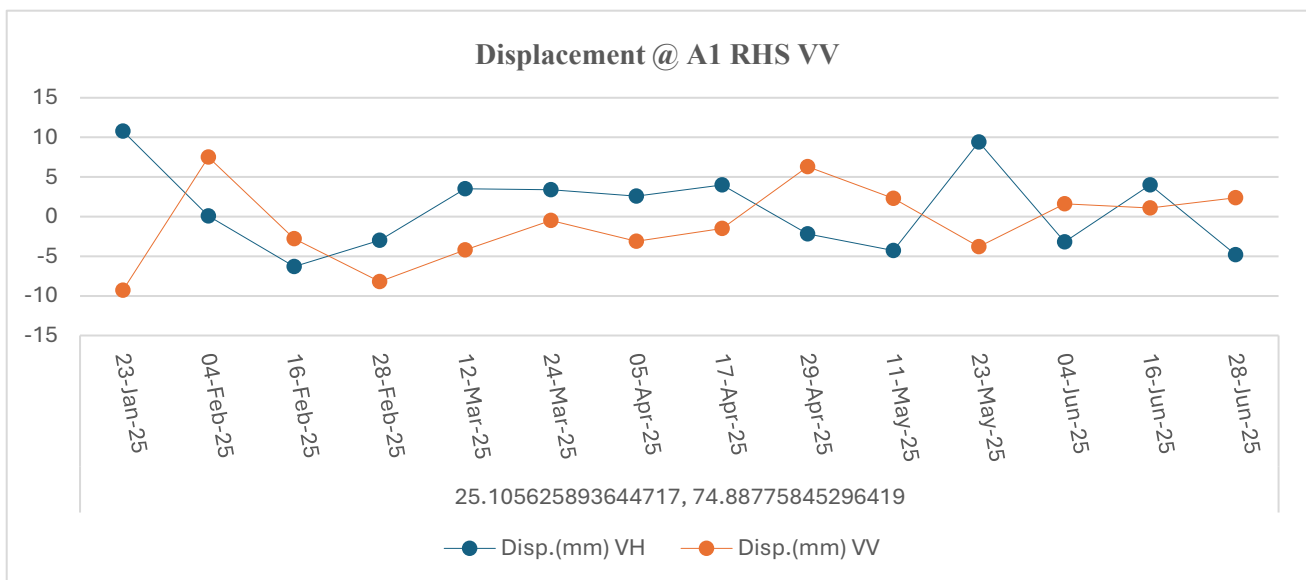
Critical Events & Anomalies:

- **Feb 2025:** VH decreases sharply to approximately –25 mm, representing the largest negative horizontal displacement in the dataset.
- **May 2025:** VV reaches one of its early peaks while VH simultaneously drops back to around –20 mm.
- **June 2025:** VV reaches its highest value, while VH settles around –20 mm, reflecting continued uplift despite persistent negative lateral shift.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.105625893644717, 74.88775845296419)



Key Observations:

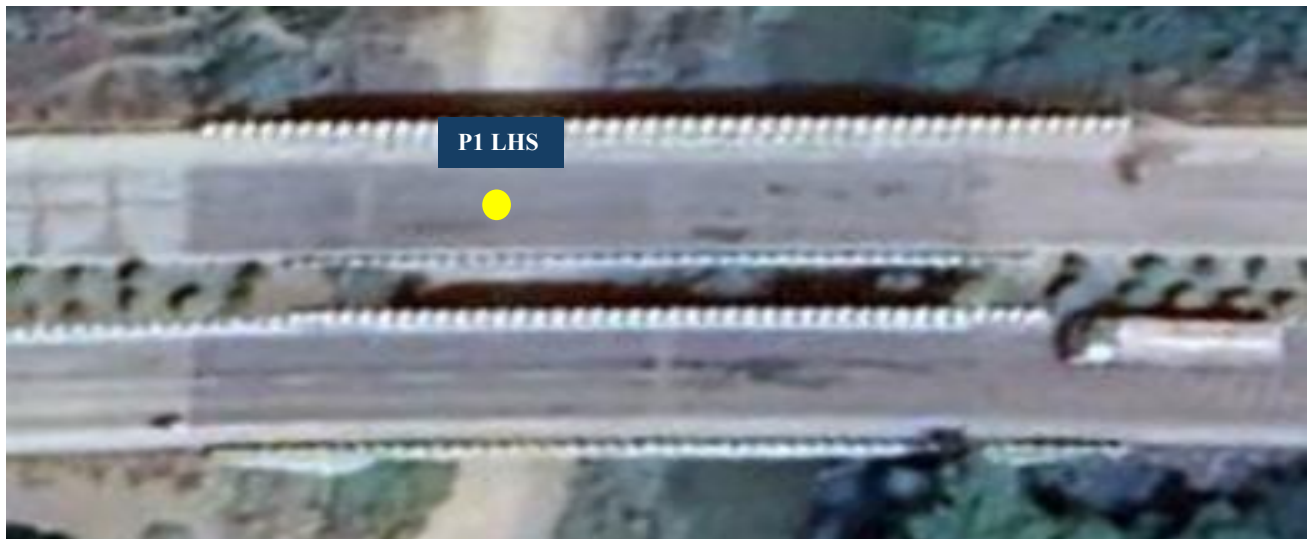
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** oscillates frequently between +10 mm and –10 mm, showing no persistent directional drift. The behaviour resembles a balanced lateral response, with alternating minor shifts but no long-term progressive movement in either direction.
- **Vertical displacement (VV)** fluctuates within a narrower band from approximately +8 mm to –10 mm, without a clear rising or falling trend.

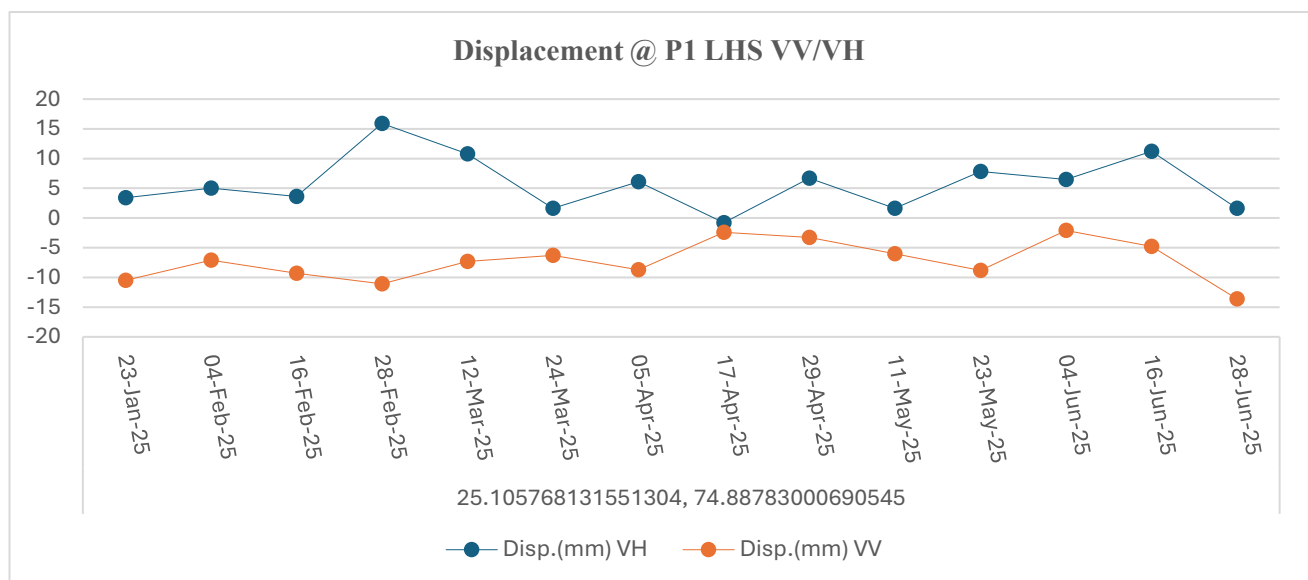
Critical Events & Anomalies:

- **January 2025:** VH begins near +12 mm, the highest recorded horizontal value, while VV starts at its lowest (–8 mm).
- **June 2025:** Both VH and VV converge near 0 to +4 mm, suggesting a return to equilibrium after months of irregular oscillations.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (25.105768131551304, 74.88783000690545)



Key Observations:

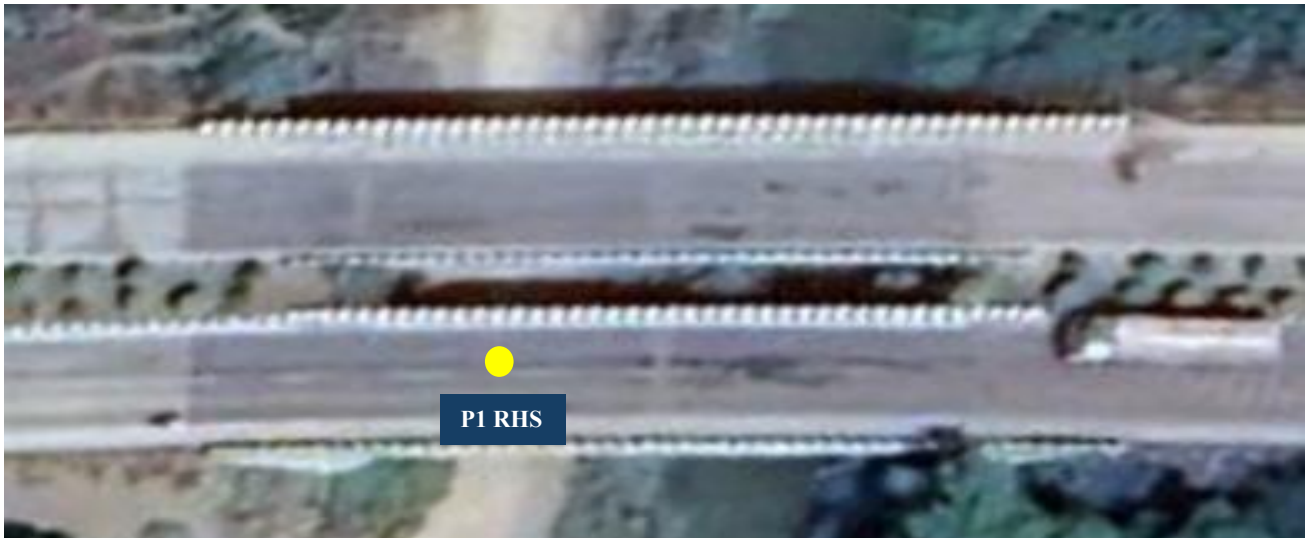
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** remains consistently negative, generally between -3 mm and -10 mm. A mild upward shift is observed during mid-March to early April, when VV improves from roughly -8 mm to -4 mm.
- **Horizontal Displacement (VH)** begins around +5 to +7 mm in January 2025 and shows a marked rise through February–March.

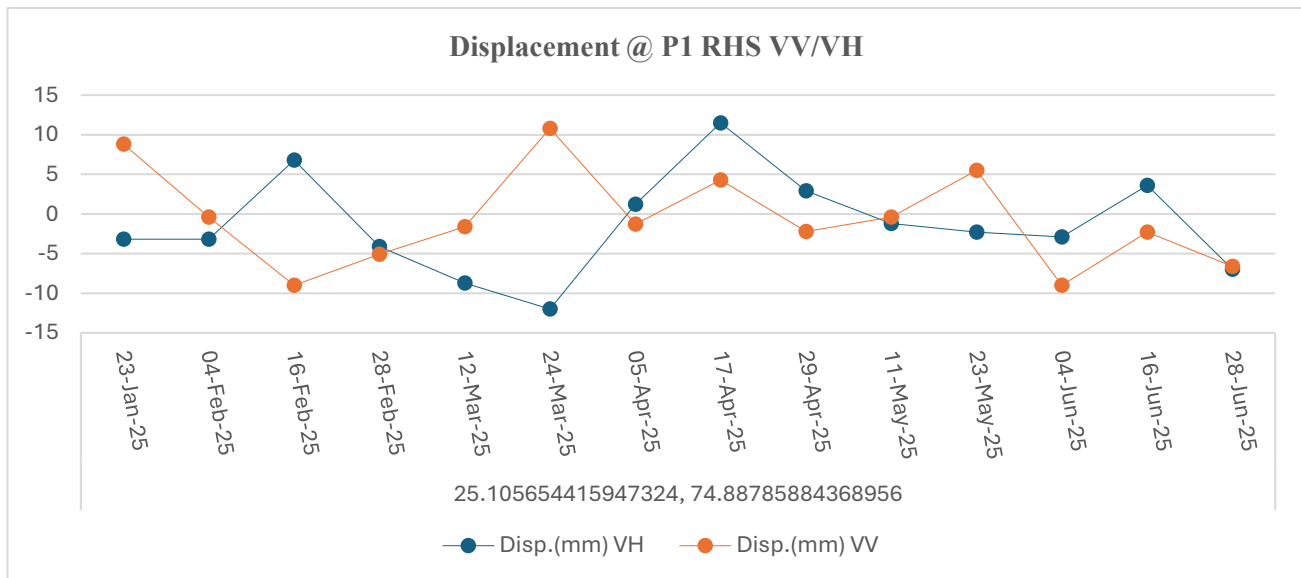
Critical Events & Anomalies:

- **March 2025:** VH exhibits a significant upward movement, peaking around +17 to +18 mm, marking the highest lateral shift in the dataset.
- **May 2025:** VH again rises noticeably (around +10 to +12 mm) after a period of lower values, showing intermittent lateral activity.
- **June 2025:** VV drops sharply to around -13 mm, representing the highest downward vertical displacement recorded, indicating a potential localized settlement event.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (25.105654415947324, 74.88785884368956)



Key Observations:

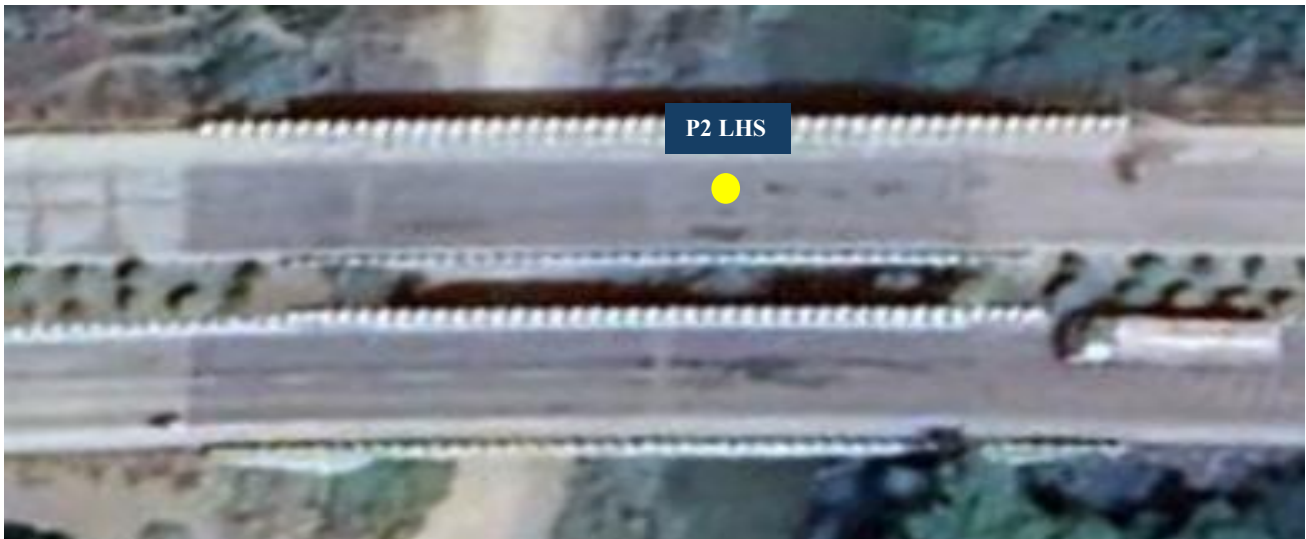
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows moderate variability, generally ranging between -10 mm to +10 mm, indicating a relatively mild vertical movement pattern with no prolonged upward or downward trend.
- **Horizontal Displacement (VH)** exhibits wider fluctuation, ranging between -12 mm to +12 mm, reflecting more dynamic lateral activity compared to the vertical component.

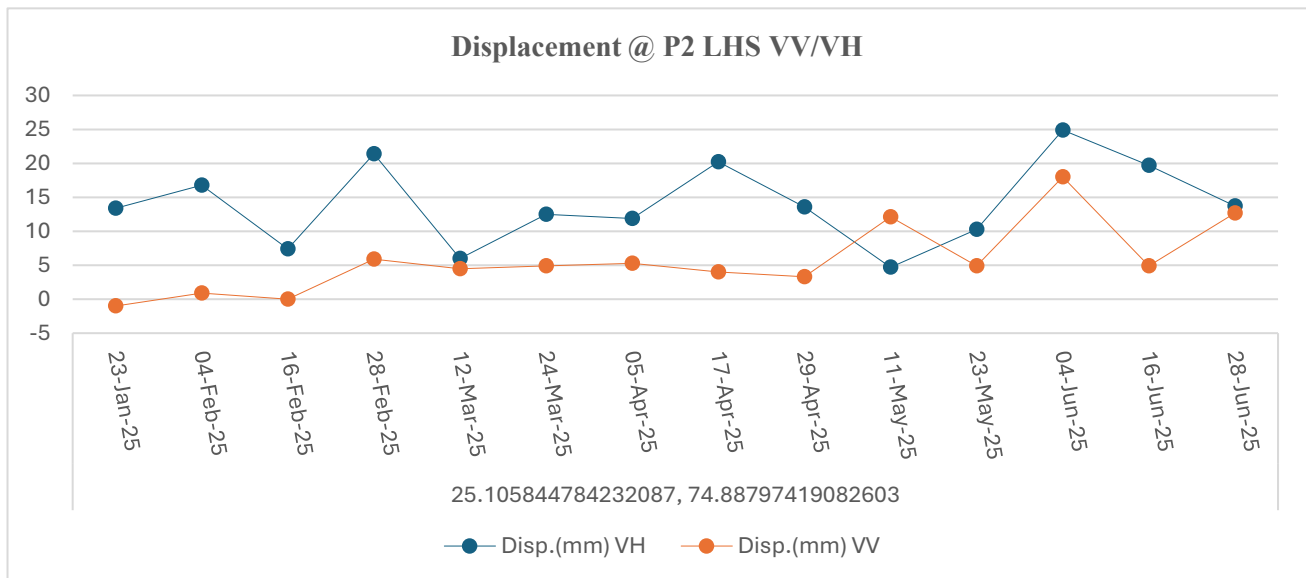
Critical Events & Anomalies:

- **January–February 2025:** VH fluctuates sharply from -5 mm to +7 mm, while VV drops to around -8 mm, indicating early instability on the horizontal axis.
- **March–April 2025:** VV spikes upward to approximately +10 mm, while VH dips to -10 mm, showing a brief period of opposite-trend displacement suggestive of structural response changes.
- **May 2025:** Both VH and VV show synchronized moderate movement (VH around +8–10 mm, VV around +5–6 mm), reflecting a short interval of coordinated displacement behaviour.

3.7 SAR Based Analysis at Pier 2 LHS:



P2 LHS (25.105844784232087, 74.88797419082603)



Key Observations:

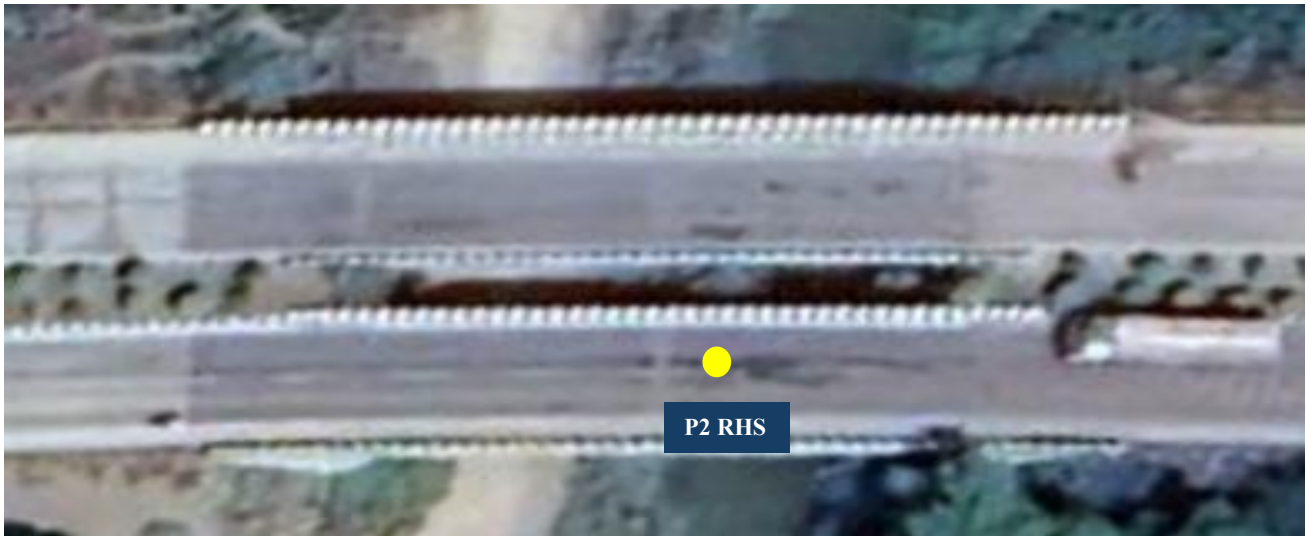
Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** shows moderate to high variability, generally ranging between 5 mm and 25 mm, with several pronounced peaks indicating intermittent lateral movement.
- **Vertical Displacement (VV)** consistently remains lower than VH, mostly fluctuating between 0 mm and 12 mm, indicating comparatively mild vertical activity at this location.

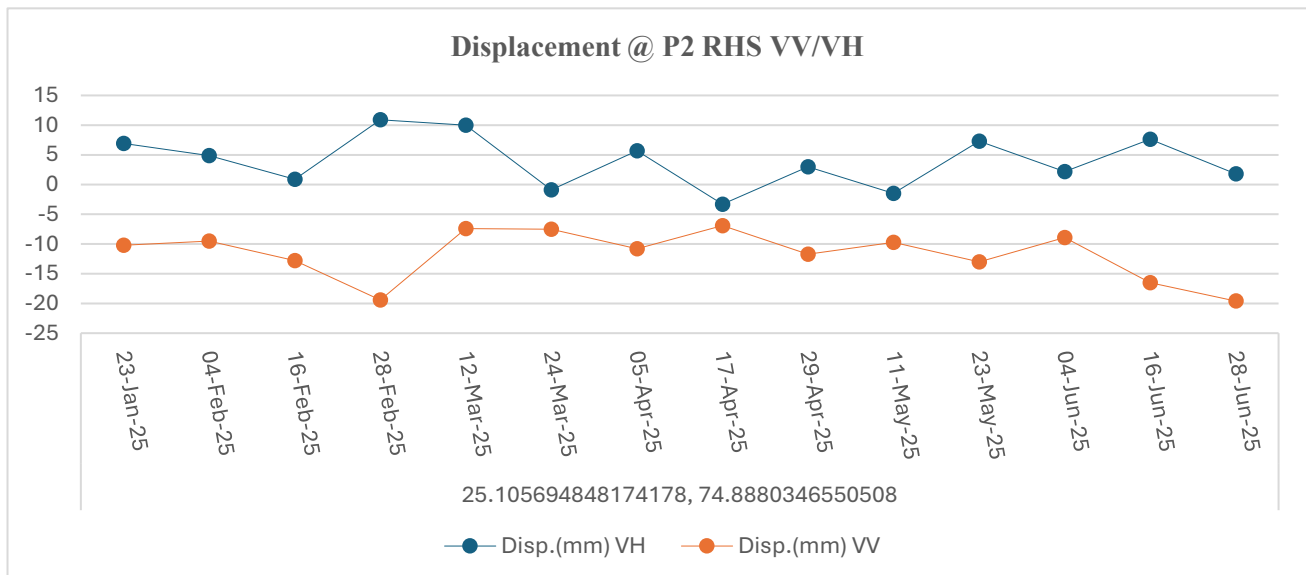
Critical Events & Anomalies:

- **February–March 2025:** VH shows sharp oscillations, rising to 20–22 mm and then dropping near 5 mm, signalling notable short-term lateral instability.
- **May 2025:** A significant VH peak of around 25 mm occurs, representing one of the highest lateral movement readings in the monitoring period.
- **June 2025:** Both VH and VV record simultaneous increases (VH 15–17 mm, VV 10–12 mm), indicating a coordinated rise in vertical and horizontal displacement that may reflect load or ground-condition changes

3.8 SAR Based Analysis at Pier 2 RHS:



P2 RHS (25.105694848174178, 74.8880346550508)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows consistently higher downward movement, mostly ranging between –10 mm to –22 mm, indicating persistent vertical settlement on the RHS of Pier P2.
- **Horizontal Displacement (VH)** fluctuates within a 5 mm to –5 mm band for most of the period, showing generally stable but oscillatory lateral behaviour, with a few positive peaks reaching 10–12 mm.

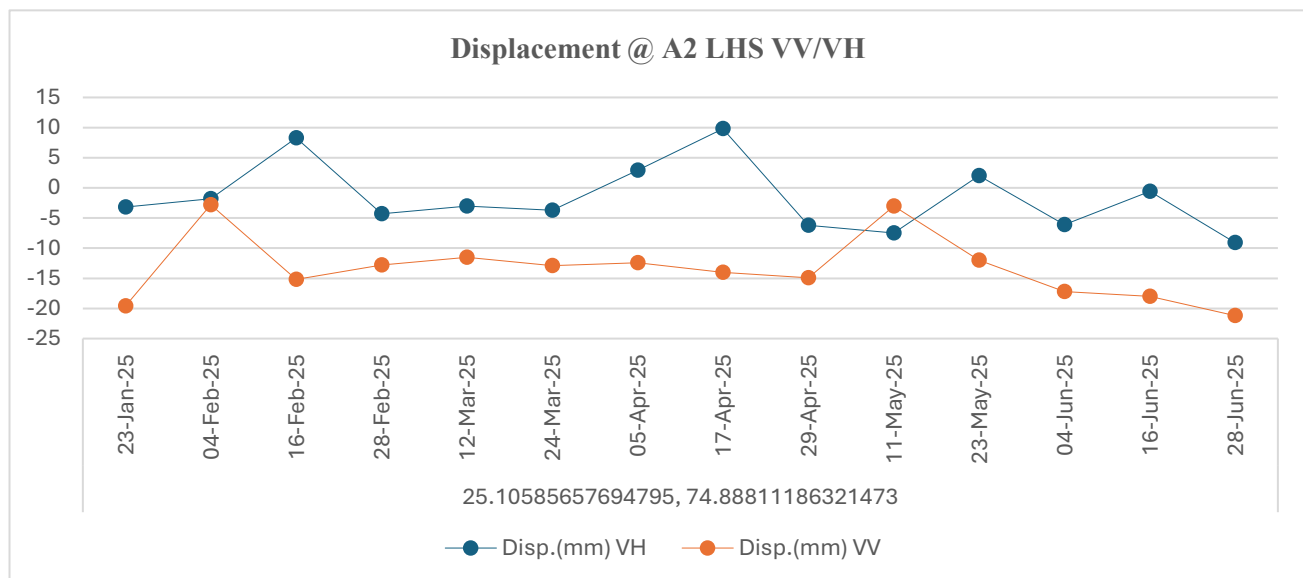
Critical Events & Anomalies:

- **February 2025:** VV drops sharply to around –20 mm, marking the most significant vertical settlement event during the monitoring period.
- **March–April 2025:** VH shows moderate recovery with values rising back to 8–10 mm, indicating short-term lateral movement after the February settlement phase.
- **May–June 2025:** VV maintains high settlement levels between –10 mm to –18 mm, while VH fluctuates between –5 mm to 10 mm, indicating continued vertical distress with alternating minor lateral shifts.

3.9 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.10585657694795, 74.88811186321473)



Key Observations:

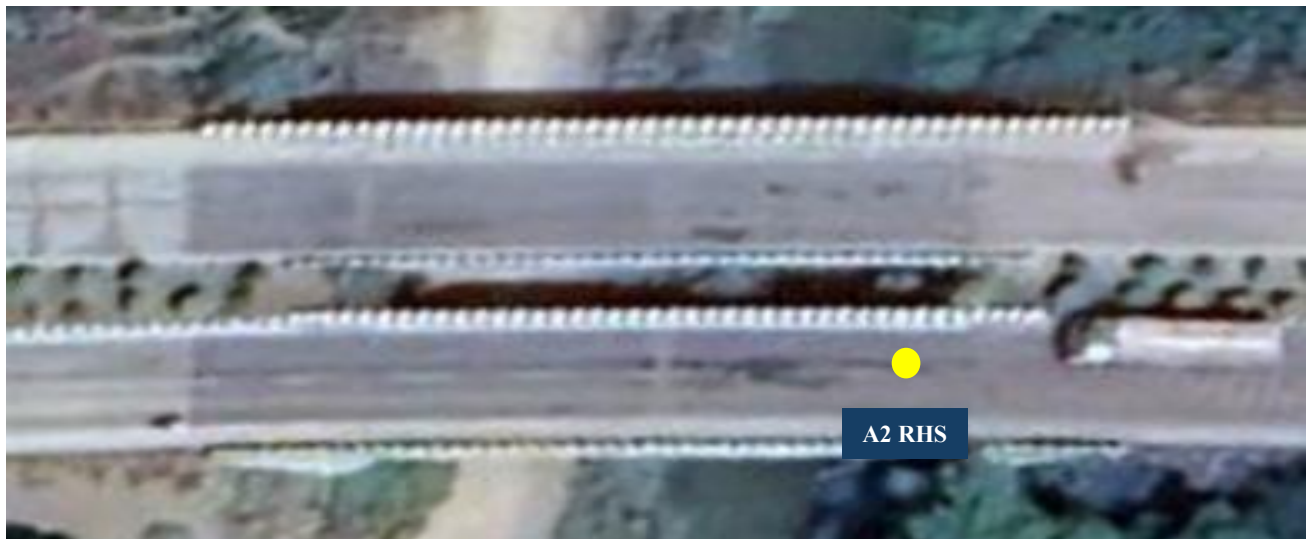
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows a predominantly downward trend, with values generally ranging between –10 mm to –22 mm, indicating consistent vertical settlement at this location.
- **Horizontal Displacement (VH)** fluctuates moderately within the –10 mm to +12 mm range. Movements remain mostly stable, with occasional short-term excursions into positive values, reflecting mild lateral variability.

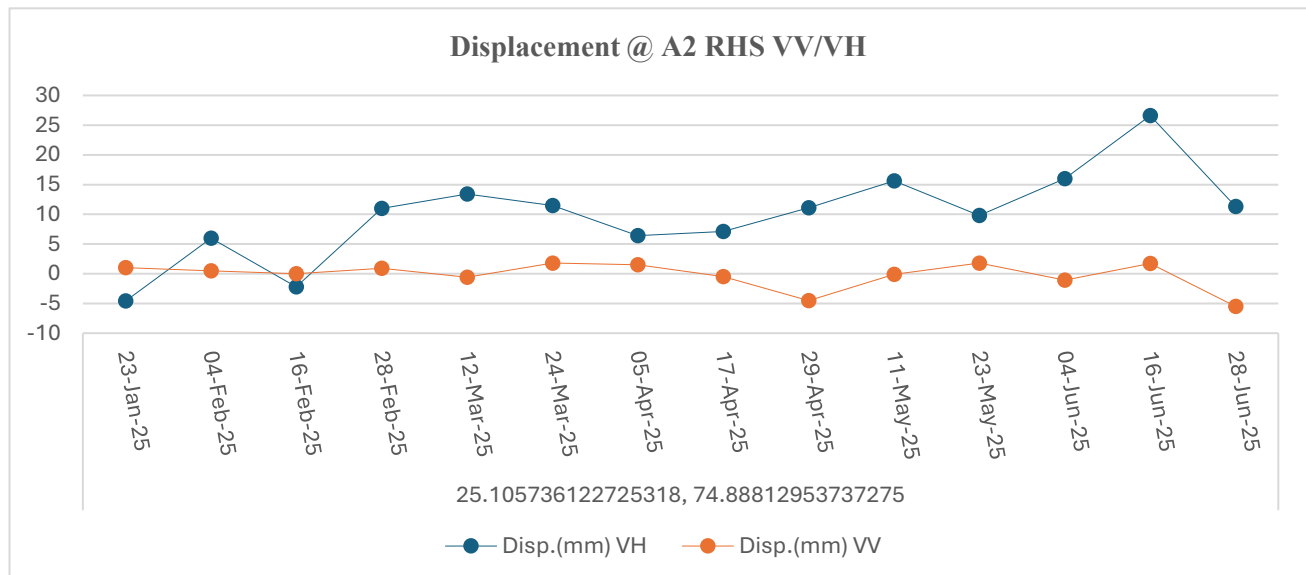
Critical Events & Anomalies:

- **February 2025:** VH peaks briefly at around +10 to +12 mm, marking a notable lateral outward movement compared to earlier readings.
- **April 2025:** Both VH and VV show significant divergence VH rises to approximately +12 mm, while VV deepens to about –12 mm, indicating a short period of combined lateral and vertical displacement activity.
- **June 2025:** VV reaches its most negative value (–22 mm), reflecting the highest vertical settlement recorded in the monitoring period.

3.10 SAR Based Analysis at Abutment 2 LHS:



A2 RHS (25.105736122725318, 74.88812953737275)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** remains within a relatively narrow negative range of approximately –12 mm to –2 mm, indicating controlled and stable vertical settlement with no significant upward rebound.
- **Horizontal Displacement (VH)** shows higher variability, ranging approximately from –5 mm to +25 mm, with several pronounced peaks indicating intermittent strong lateral movement.

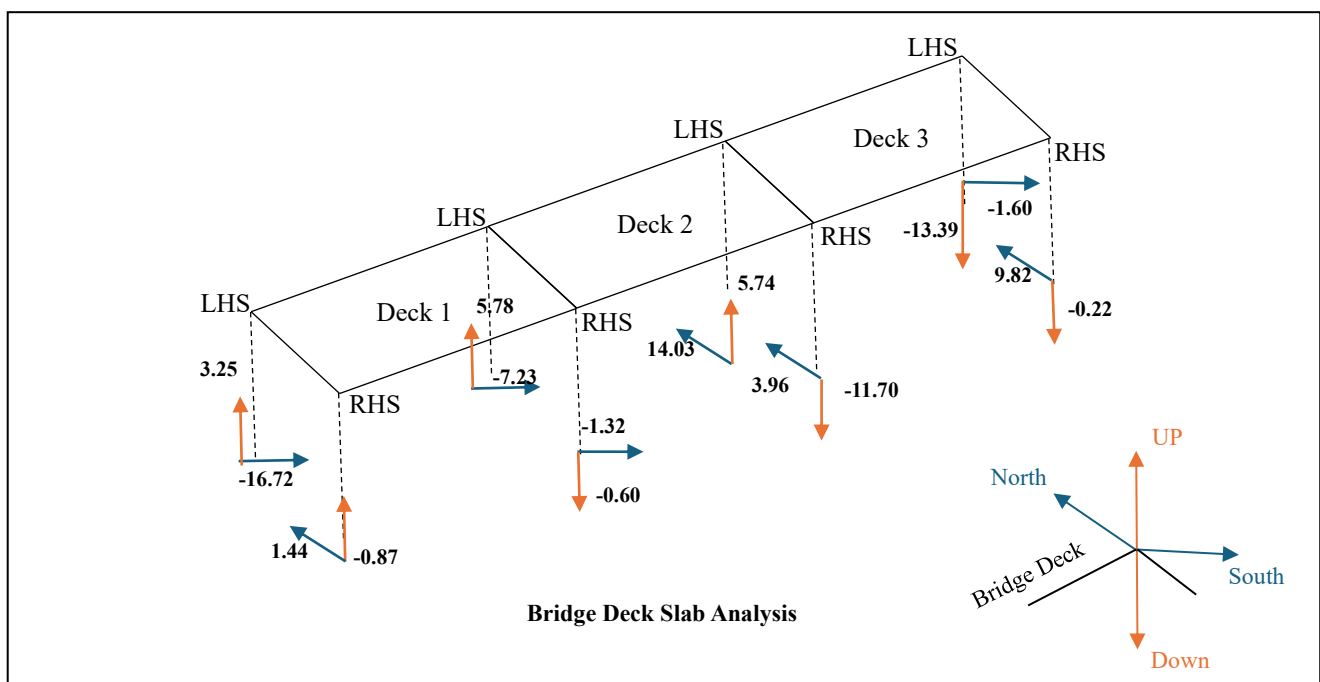
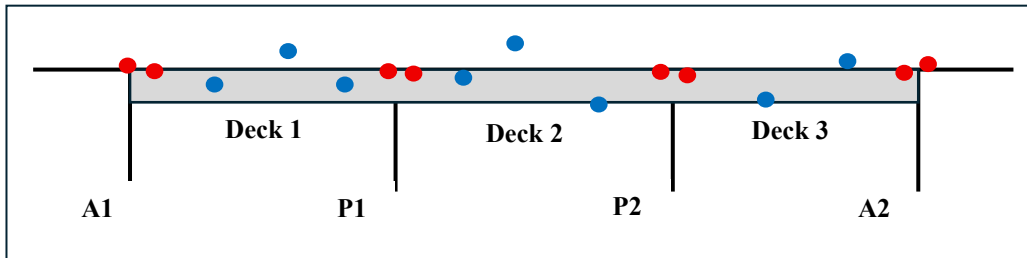
Critical Events & Anomalies:

- **February–March 2025:** VH shows a rising trend, peaking between 10–15 mm, suggesting an early period of increased lateral displacement.
- **June 2025:** VV shows a downward shift to around –8 mm, indicating a brief period of increased downward vertical displacement while VH remains elevated.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- **Points of Analysis (POA) of bridge (●)**



Vertical Displacement Behaviour:

- The bridge deck shows significant downward vertical displacement, with values ranging from approximately -16.72 mm to -13.39 mm at several RHS locations, indicating stronger settlement or sagging toward the right-hand side.

Horizontal Displacement Behaviour:

- Horizontal displacements at the RHS show a mix of eastward and south-eastward components, with magnitudes ranging approximately from 0.87 mm to 14.03 mm, suggesting outward drift away from the bridge centreline.

CONCLUSION:

- The displacement field across Decks 1, 2, and 3 indicates a combined vertical sagging (RHS) and horizontal divergence (LHS vs RHS), characteristic of torsional deformation developing across the deck slab. The magnitude differences between left and right sides, together with opposite directional shifts, confirm that the deck is not undergoing simple settlement but exhibiting warping behaviour.